

# Assessment (Week 4)

## Description

**Module/Week4:** Assignment 3

**Topic:** Incremental Refresh in Data Warehousing

*Student's Full Name: Vishal Guddeti*

*Course Title: Data Warehousing and Analytics in the Cloud*

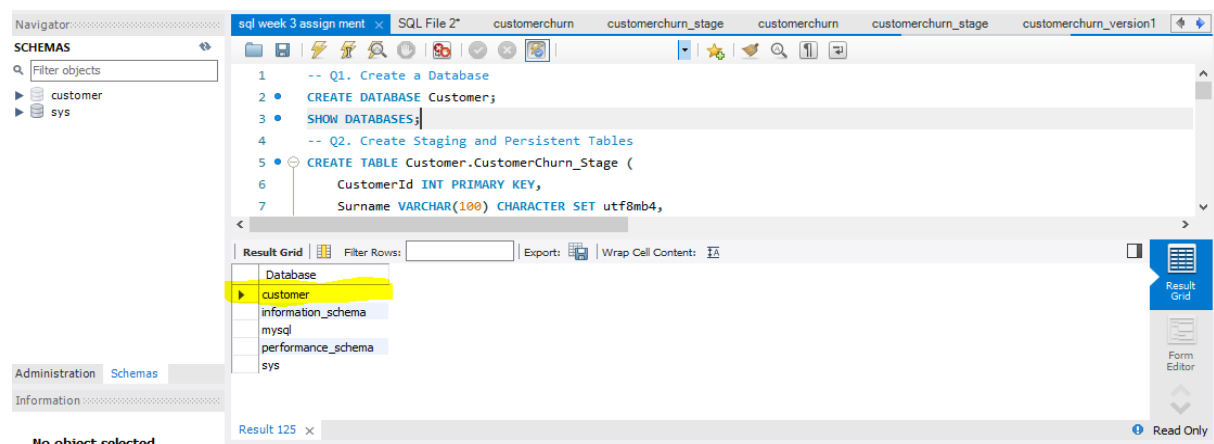
*Term name and year: Fall 2023*

*Submission Week: Week 4 - Assignment 3*

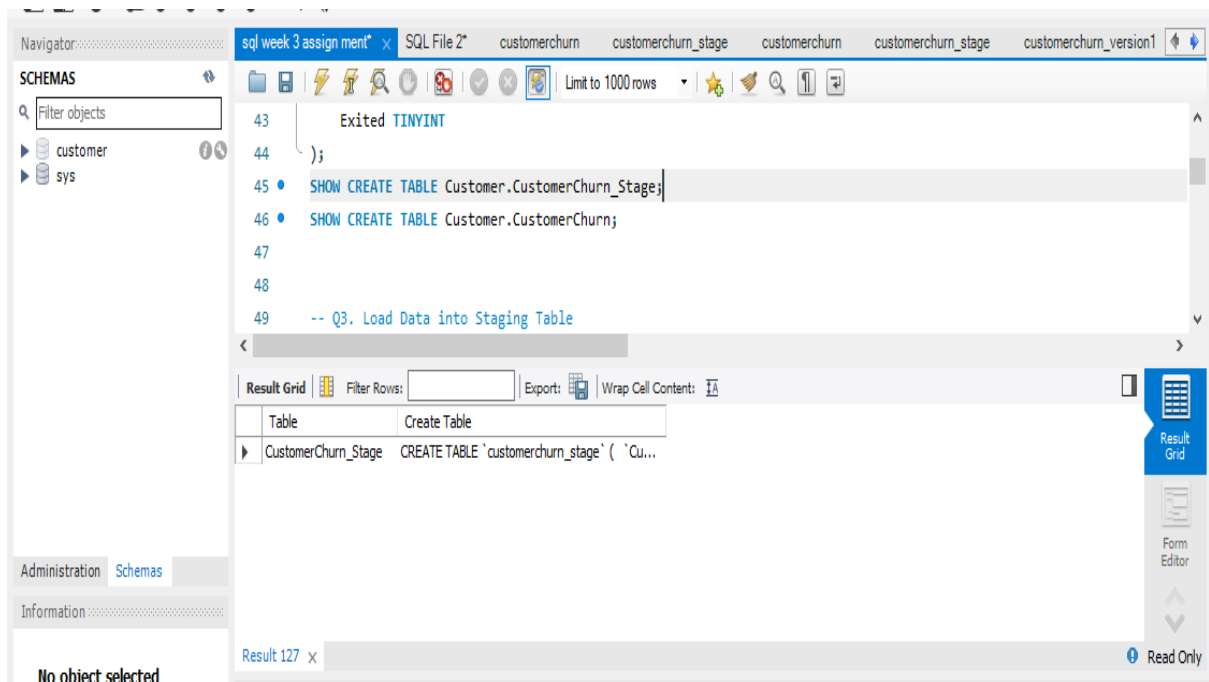
*Instructor's Name: Dr.Nayem Rahman*

*Date of Submission:07-08-2024*

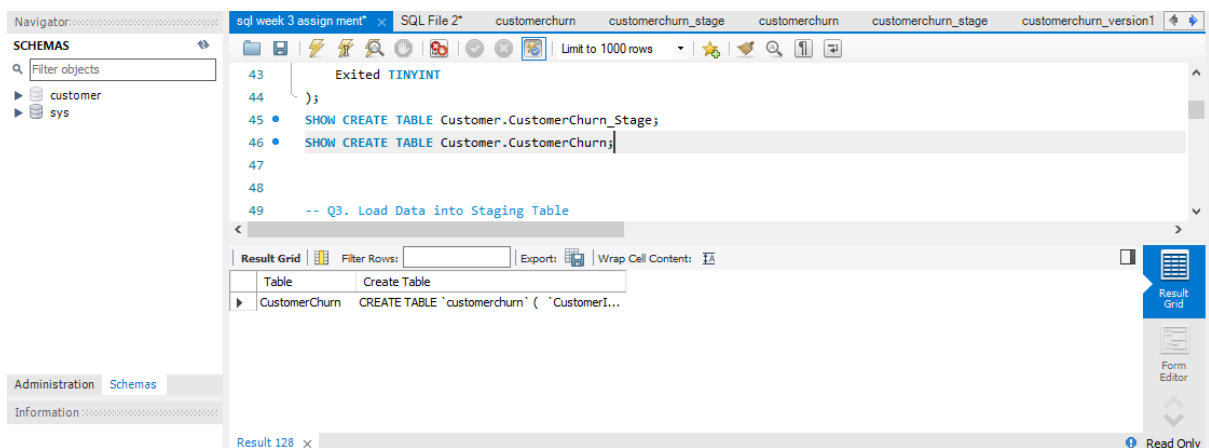
**Q1. { A } Using the MySQL Workbench, create a database called Customer. The database must be named "Customer". { B } Check if the database was created and use the same for further questions**



**Q2. { A } Create a staging table, `** Customer.CustomerChurn_Stage **`, in a database system, with the column list provided in the CSV file. Define the 'CustomerId' as the Primary Key (PK). Get the table definition (DDL) from the database system and capture it in a Word document for submission. { B } Create a persistent table, `** Customer.CustomerChurn **`, with the column list provided in the CSV file + following 5 columns : << SourceSystemNm NVARCHAR(20) NOT NULL , CreateAgentId NVARCHAR(20) NOT NULL , CreateDtm DATETIME NOT NULL, ChangeAgentId NVARCHAR(20) NOT NULL , ChangeDtm DATETIME NOT NULL >> Define the 'CustomerId' as the Primary Key (PK). Get the table definition (DDL) from the database system and capture it in a Word document for submission.**



**B.**



**Q3. { A } Load the staging table, `** Customer.CustomerChurn_Stage **`, with data from the CSV file, CustomerChurn1.csv . { B } Verify data by comparing the row counts between the CSV file and the staging table, `** Customer.CustomerChurn_Stage [Data Source: CustomerChurn1.CSV] **`. Provide the screenshot of last few rows using the 'SELECT \* ' . Make sure the output shows all column values. The SELECT statement must use the ORDER BY 'CustomerId' .**

Navigator: sql week 3 assign ment SQL File 2\* customerchurn customerchurn\_stage customerchurn customerchurn\_stage customerchurn\_version1

Limit to 1000 rows

```

58 • SELECT COUNT(*) FROM Customer.CustomerChurn_Stage;
59 • SELECT * FROM Customer.CustomerChurn_Stage
60 ORDER BY CustomerId
61 LIMIT 10 OFFSET 90;
62
63 -- Q4. Create Stored Procedure
64 DELIMITER //

```

Result Grid

| CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited |
|------------|-----------|-------------|-----------|--------|-----|-----------|--------|
| 15780961   | Cavenagh  | 735         | France    | Female | 21  | 178718.19 | 0      |
| 15788218   | Henderson | 549         | Spain     | Female | 24  | 0.00      | 0      |
| 15788448   | Watson    | 490         | Spain     | Male   | 31  | 145260.23 | 0      |
| 15789484   | Hammond   | 751         | Germany   | Female | 36  | 169831.46 | 0      |
| 15792365   | He        | 501         | France    | Male   | 44  | 142051.07 | 0      |
| 15794171   | Lombardo  | 475         | France    | Female | 45  | 134264.04 | 1      |
| 15803136   | Postle    | 416         | Germany   | Female | 41  | 122189.66 | 0      |
| 15804771   | Velazquez | 614         | France    | Male   | 51  | 40685.92  | 0      |

Administration Schemas

Information

CustomerChurn\_Stage 129 x

Apply

**Q5. Execute the stored procedure, \*\* Customer.PrCustomerChurn \*\*, that was created in Q4. After execution, the stored procedure should load data from the stage to the persistent table: \*\* Customer.CustomerChurn \*\*. {A} Verify data by comparing the row counts between the staging table, \*\* Customer.CustomerChurn\_Stage [Data Source: CustomerChurn1.CSV] \*\* and the persistent table: \*\* Customer.CustomerChurn \*\*. { B } Provide the screenshot of last few rows using the SELECT \*. Make sure the output shows all column values. The SELECT statement must use the ORDER BY CustomerId.**

File Edit View Query Database Server Tools Scripting Help

Navigator: sql week 3 assign ment SQL File 2\* customerchurn customerchurn\_stage customerchurn customerchurn\_stage customerchurn\_version1

Limit to 1000 rows

```

134 CALL Customer.PrCustomerChurn();
135
136 SELECT COUNT(*) AS StagingRowCount FROM Customer.CustomerChurn_Stage;
137 SELECT COUNT(*) AS PersistentRowCount FROM Customer.CustomerChurn;
138
139 SELECT * FROM Customer.CustomerChurn ORDER BY CustomerId DESC LIMIT 10;
140 -- Q6. Create and Populate Version Table

```

Result Grid

| CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId | CreateDtm           |
|------------|-----------|-------------|-----------|--------|-----|-----------|--------|----------------|---------------|---------------------|
| 16812518   | Palermo   | 657         | Spain     | Female | 37  | 163607.18 | 0      | CSV            | Admin         | 2024-08-07 19:50:33 |
| 15809248   | Cole      | 524         | France    | Female | 36  | 0.00      | 0      | CSV            | Admin         | 2024-08-07 19:50:32 |
| 15805254   | Ndukaku   | 652         | Spain     | Female | 75  | 0.00      | 0      | CSV            | Admin         | 2024-08-07 19:50:32 |
| 15804771   | Velazquez | 614         | France    | Male   | 51  | 40685.92  | 0      | CSV            | Admin         | 2024-08-07 19:50:32 |
| 15803136   | Postle    | 416         | Germany   | Female | 41  | 122189.66 | 0      | CSV            | Admin         | 2024-08-07 19:50:32 |
| 15794171   | Lombardo  | 475         | France    | Female | 45  | 134264.04 | 1      | CSV            | Admin         | 2024-08-07 19:50:32 |
| 15792365   | He        | 501         | France    | Male   | 44  | 142051.07 | 0      | CSV            | Admin         | 2024-08-07 19:50:32 |

Administration Schemas

Information

No object selected

Result 150 Result 151 Result 152 x

Read Only

Navigator: sql week 3 assign ment\* SQL File 2\* assment3\* customerchurn\_stage customer customerchurn\_stage

SCHEMAS

Filter objects

customer

Tables

customerchurn

customerchurn\_stage

customerchurn\_version1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Table: customerchurn\_stage

```

126 • SELECT COUNT(*) AS PersistentCount FROM Customer.CustomerChurn;
127
128 • SELECT * FROM Customer.CustomerChurn ORDER BY CustomerId DESC LIMIT 10;
129
130 -- Q6
131 • CREATE TABLE Customer.CustomerChurn_Version1 AS
132

```

Result Grid

|   | Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId  | CreateDtm           | ChangeAgentId  | ChangeDtm           |
|---|-----------|--------|-----|-----------|--------|----------------|----------------|---------------------|----------------|---------------------|
| ▶ | Spain     | Female | 37  | 163607.18 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
|   | France    | Female | 36  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
|   | Spain     | Female | 75  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
|   | France    | Male   | 51  | 40685.92  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
|   | Germany   | Female | 41  | 122189.66 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
|   | France    | Female | 45  | 134264.04 | 1      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
|   | France    | Male   | 44  | 147051.07 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |

CustomerChurn 49 x

Q6. After data verification is completed, in Q5 , { A } create table, \*\* Customer.CustomerChurn\_Version1 \*\*, with data from \*\* Customer.CustomerChurn \*\* (that was already loaded from Customer.CustomerChurn\_Stage via the stored procedure). { B } Show table definition of Customer.CustomerChurn\_Version1 and show the row count of the table, \*\* Customer.CustomerChurn\_Version1 \*\*: { C } Provide the screenshot of last few rows for \*\* Customer.CustomerChurn\_Version1 \*\* [Originally data came from: CustomerChurn1.CSV]. Make sure the output shows all column values. The SELECT statement must use the ORDER BY CustomerId. { D } Empty the staging table, \*\* Customer.CustomerChurn\_Stage \*\*, and load it with data from the CSV file, "CustomerChurn2.csv ". Verify data by comparing the row counts between the CSV file and the staging table, \*\* Customer.CustomerChurn\_Stage \*\* [Data Source: CustomerChurn2.CSV]. Provide the row count of \*\* Customer.CustomerChurn\_Stage \*\* that you loaded from CustomerChurn2.csv file. Provide the screenshot of last few rows using the SELECT \*. Make sure the output shows all column values. The SELECT statement must use the ORDER BY CustomerId.

## Customer.CustomerChurn\_Version1

Navigator: sql week 3 assign ment\* SQL File 2\* assment3 customerchurn\_stage

SCHEMAS

Filter objects

customer

Tables

customerchurn

customerchurn\_stage

customerchurn\_version1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

```

144 SHOW CREATE TABLE Customer.CustomerChurn_Version1;
145 SELECT COUNT(*) AS Version1RowCount FROM Customer.CustomerChurn_Version1;
146
147 SELECT * FROM Customer.CustomerChurn_Version1 ORDER BY CustomerId DESC LIMIT 10;
148
149 TRUNCATE TABLE Customer.CustomerChurn_Stage;
150

```

Result Grid

|   | CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId | CreateDtm           |
|---|------------|-----------|-------------|-----------|--------|-----|-----------|--------|----------------|---------------|---------------------|
| ▶ | 16812518   | Palermo   | 657         | Spain     | Female | 37  | 163607.18 | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15809248   | Cole      | 524         | France    | Female | 36  | 0.00      | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15805254   | Ndukaku   | 652         | Spain     | Female | 75  | 0.00      | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15804771   | Velazquez | 614         | France    | Male   | 51  | 40685.92  | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15803136   | Postle    | 416         | Germany   | Female | 41  | 122189.66 | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15794171   | Lombardo  | 475         | France    | Female | 45  | 134264.04 | 1      | CSV            | Admin         | 2024-08-07 21:49:44 |
|   | 15792365   | He        | 501         | France    | Male   | 44  | 147051.07 | 0      | CSV            | Admin         | 2024-08-07 21:49:44 |

Result 242 Result 243 Result 244 x

Read Only

Navigator: sql week 3 assign ment SQL File 2 assment3 customerchurn\_stage customer customerchurn\_stage

Limit to 1000 rows

138

139 • `SELECT COUNT(*) AS Version1RowCount FROM Customer.CustomerChurn_Version1;`

140

141

142

143 • `SELECT * FROM Customer.CustomerChurn_Version1 ORDER BY CustomerId DESC LIMIT 10;`

144

Result Grid

| Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId  | CreateDtm           | ChangeAgentId  | ChangeDtm           |
|-----------|--------|-----|-----------|--------|----------------|----------------|---------------------|----------------|---------------------|
| Spain     | Female | 37  | 163607.18 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| France    | Female | 36  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| Spain     | Female | 75  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| France    | Male   | 51  | 40685.92  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| Germany   | Female | 41  | 122189.66 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| France    | Female | 45  | 134264.04 | 1      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |
| France    | Male   | 44  | 142051.07 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root@localhost | 2024-08-07 22:59:18 |

CustomerChurn\_Version1 50

## Customer.CustomerChurn\_Stage

Navigator: Schemas

Filter objects

customer

Tables

customerchurn

customerchurn\_stage

customerchurn\_version1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Table:

149 -- Show row count of the staging table

150 • `SELECT COUNT(*) AS RowCount FROM Customer.CustomerChurn_Stage;`

151

152 -- Display last few rows from the staging table

153 • `SELECT * FROM Customer.CustomerChurn_Stage ORDER BY CustomerId DESC LIMIT 10;`

154

155

Result Grid

| CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited |
|------------|-----------|-------------|-----------|--------|-----|-----------|--------|
| 16812518   | Palermo   | 657         | Spain     | Female | 37  | 163607.18 | 0      |
| 15809248   | Cole      | 524         | France    | Female | 36  | 0.00      | 0      |
| 15805254   | Ndukaku   | 652         | Spain     | Female | 75  | 0.00      | 0      |
| 15804771   | Velazquez | 614         | France    | Male   | 51  | 40685.92  | 0      |
| 15803136   | Postle    | 416         | Germany   | Female | 41  | 122189.66 | 0      |
| 15794171   | Lombardo  | 475         | France    | Female | 45  | 134264.04 | 1      |
| 15792365   | He        | 501         | France    | Male   | 44  | 142051.07 | 0      |
| 15780484   | Hammond   | 751         | Germany   | Female | 36  | 163607.18 | 0      |

CustomerChurn\_Stage 56

**Q7. Execute the stored procedure, Customer.PrCustomerChurn, that was created in Q4. After execution, the stored procedure should load data from the stage to the persistent table: Customer.CustomerChurn. CALL `customer`.`PrCustomerChurn`(); This time, the table will be refreshed via DELETE, UPDATE, and INSERT/SELECT statements in the stored procedure. Show the row count results of both Customer.CustomerChurn\_Version1 table [Data Source: CustomerChurn1.CSV] and the persistent table: Customer.CustomerChurn. Compare the rows between the Customer.CustomerChurn\_Version1 [Data Source: CustomerChurn1.CSV] table and the persistent table: Customer.CustomerChurn [Data Source: CustomerChurn2.CSV]. Show the rows that are available in the Customer.CustomerChurn\_Version1 table but not in the Customer.CustomerChurn table (implementation of brand-new row DELETE statement of the stored procedure).**

**SCHEMAS**

Filter objects

- customer
  - Tables
    - customerchurn
    - customerchurn\_stage
    - customerchurn\_version1
  - Views
  - Stored Procedures
  - Functions
- sys

Administration Schemas

Information

Table: customerchurn\_stage

Limit to 1000 rows

```

170 -- Show rows that are in Customer.CustomerChurn_Version1 but not in Customer.CustomerChurn
171 • SELECT *
172 FROM Customer.CustomerChurn_Version1
173 WHERE CustomerId NOT IN (SELECT CustomerId FROM Customer.CustomerChurn);
174
175 -- Q8
176 • SELECT

```

Result Grid

| CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId  | CreateDtm           | Ch   |
|------------|-----------|-------------|-----------|--------|-----|-----------|--------|----------------|----------------|---------------------|------|
| 15604348   | Allard    | 710         | Spain     | Male   | 22  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |
| 15687946   | Osborne   | 556         | France    | Female | 61  | 117419.35 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |
| 15701164   | Onyeorulu | 506         | France    | Female | 34  | 90307.62  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |
| 15725737   | Mosman    | 669         | France    | Male   | 46  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |
| 15755648   | Pisano    | 675         | France    | Female | 21  | 98373.26  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |
| 15812518   | Palermo   | 657         | Spain     | Female | 37  | 163607.18 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 22:59:18 | root |

CustomerChurn\_Version1 57

Read Only

**Q8. Show the rows (SELECT \*) that changed (one or many non-Primary Key columns), in the Customer.CustomerChurn table (implementation of UPDATE statement of the stored procedure). You need to perform a comparison between Customer.CustomerChurn table [Data Source: CustomerChurn2.CSV] and Customer.CustomerChurn\_Version1 table [Data Source: CustomerChurn1.CSV] in terms of non-PK columns (Excludes: SourceSystemNm, CreateAgentId, CreateDtm, ChangeAgentId, ChangeDtm), and with a join condition using the PK column(s). You must do ORDER BY CustomerId. The output of this query should show different values for the CreateDtm and ChangeDtm columns in Customer.CustomerChurn table for the changed rows. Take a screenshot and capture it in the Word document. Make sure all columns including CreateDtm and ChangeDtm of CustomerChurn table are displayed.**

**SCHEMAS**

Filter objects

- customer
  - Tables
    - customerchurn
    - customerchurn\_stage
    - customerchurn\_version1
  - Views
  - Stored Procedures
  - Functions
- sys

Administration Schemas

Information

Table: customerchurn\_stage

Limit to 1000 rows

```

202 OR ChurnTbl.Balance <> Version1Tbl.Balance
203 OR ChurnTbl.Exited <> Version1Tbl.Exited
204 ORDER BY ChurnTbl.CustomerId;
205
206
207 -- Q9
208 -- Execute the query to find rows in Customer.CustomerChurn but not in Customer.CustomerChurn_Version1

```

Result Grid

| CustomerId | ChurnSurname | ChurnCreditScore | ChurnGeography | ChurnGender | ChurnAge | ChurnBalance | ChurnExited | Version1Surname | Version1CreditScore | Version1Geography | Version1Gender | Version1Age | Version1Balance | Version1Exited | ChurnCreatedDtm     | ChurnChangedDtm     |
|------------|--------------|------------------|----------------|-------------|----------|--------------|-------------|-----------------|---------------------|-------------------|----------------|-------------|-----------------|----------------|---------------------|---------------------|
| 15592389   | Vivek        | 684              | India          | Male        | 27       | 134603.88    | 0           | H?              | 684                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15597945   | Dellucci     | 700              | Spain          | Female      | 32       | 654.00       | 0           | Dellucci        | 636                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15683553   | Osman        | 788              | USA            | Male        | 22       | 75888.30     | 0           | O'Brien         | 788                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15691483   | Chin         | 549              | France         | Female      | 25       | 12345.50     | 0           | Chin            | 549                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15737173   | Andrews      | 497              | Spain          | Male        | 30       | 0.00         | 1           | Andrews         | 497                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15738751   | Beit         | 493              | France         | Female      | 46       | 321.00       | 0           | Beit            | 493                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 15767821   | Sharon       | 528              | Honn Kinn      | Male        | 31       | 107016.72    | 0           | Bearne          | 528                 |                   |                |             |                 |                | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |

Result 60

Read Only

**SCHEMAS**

Filter objects

- customer
  - Tables
    - customerchurn
    - customerchurn\_stage
    - customerchurn\_version1
  - Views
  - Stored Procedures
  - Functions
- sys

Administration Schemas

Information

Table: customerchurn\_stage

Limit to 1000 rows

```

202 OR ChurnTbl.Balance <> Version1Tbl.Balance
203 OR ChurnTbl.Exited <> Version1Tbl.Exited
204 ORDER BY ChurnTbl.CustomerId;
205
206
207 -- Q9
208 -- Execute the query to find rows in Customer.CustomerChurn but not in Customer.CustomerChurn_Version1

```

Result Grid

| Version1CreditScore | Version1Geography | Version1Gender | Version1Age | Version1Balance | Version1Exited | ChurnCreatedDtm     | ChurnChangedDtm     |
|---------------------|-------------------|----------------|-------------|-----------------|----------------|---------------------|---------------------|
| 684                 | France            | Male           | 27          | 134603.88       | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 636                 | Spain             | Female         | 32          | 0.00            | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 788                 | France            | Female         | 33          | 0.00            | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 549                 | France            | Female         | 25          | 0.00            | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 497                 | Spain             | Male           | 24          | 0.00            | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 493                 | France            | Female         | 46          | 0.00            | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |
| 528                 | France            | Male           | 31          | 107016.72       | 0              | 2024-08-07 22:59:18 | 2024-08-07 23:02:11 |

Read Only

**Q9. Provide the screenshot of last few rows using the `SELECT * FROM Customer.CustomerChurn`. Make sure the output shows all column values. The `SELECT` statement must use the `ORDER BY CustomerId`. Show the rows that are available in the `Customer.CustomerChurn` table [Data Source: `CustomerChurn2.CSV`] but not in the `Customer.CustomerChurn_Version1` table (implementation of brand-new rows `INSERT` by the stored procedure). Do a `SELECT *` along with `ORDER BY CustomerId`. Take a screenshot and capture it in the Word document.**

208 -- Execute the query to find rows in Customer.CustomerChurn but not in Customer.CustomerChurn\_Version1

209

210 • `SELECT *`

211 `FROM Customer.CustomerChurn`

212 `WHERE CustomerId NOT IN (SELECT CustomerId FROM Customer.CustomerChurn_Version1)`

213 `ORDER BY CustomerId DESC LIMIT 10;`

214

| CustomerId | Surname   | CreditScore | Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId  | CreateDtm           |
|------------|-----------|-------------|-----------|--------|-----|-----------|--------|----------------|----------------|---------------------|
| 16812518   | Palermo   | 657         | Spain     | Female | 37  | 163607.18 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15771977   | Tao       | 730         | France    | Female | 39  | 99010.67  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15727556   | O'Donnell | 744         | Spain     | Female | 26  | 166297.89 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15726676   | Marshall  | 616         | Spain     | Male   | 30  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15698932   | Groves    | 756         | Germany   | Male   | 44  | 137452.09 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15657566   | Wleick    | 634         | Germany   | Male   | 24  | 103097.85 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |
| 15590975   | Markan    | 646         | France    | Female | 73  | 97759.75  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 |

208 -- Execute the query to find rows in Customer.CustomerChurn but not in Customer.CustomerChurn\_Version1

209

210 • `SELECT *`

211 `FROM Customer.CustomerChurn`

212 `WHERE CustomerId NOT IN (SELECT CustomerId FROM Customer.CustomerChurn_Version1)`

213 `ORDER BY CustomerId DESC LIMIT 10;`

214

| Geography | Gender | Age | Balance   | Exited | SourceSystemNm | CreateAgentId  | CreateDtm           | ChangeAgentId  | ChangeDtm           |
|-----------|--------|-----|-----------|--------|----------------|----------------|---------------------|----------------|---------------------|
| Spain     | Female | 37  | 163607.18 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| France    | Female | 39  | 99010.67  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| Spain     | Female | 26  | 166297.89 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| Spain     | Male   | 30  | 0.00      | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| Germany   | Male   | 44  | 137452.09 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| Germany   | Male   | 24  | 103097.85 | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |
| France    | Female | 73  | 97759.75  | 0      | Kaggle-CSV     | root@localhost | 2024-08-07 23:02:10 | root@localhost | 2024-08-07 23:02:10 |